AI Challenges in Intellectual Property

Part 2

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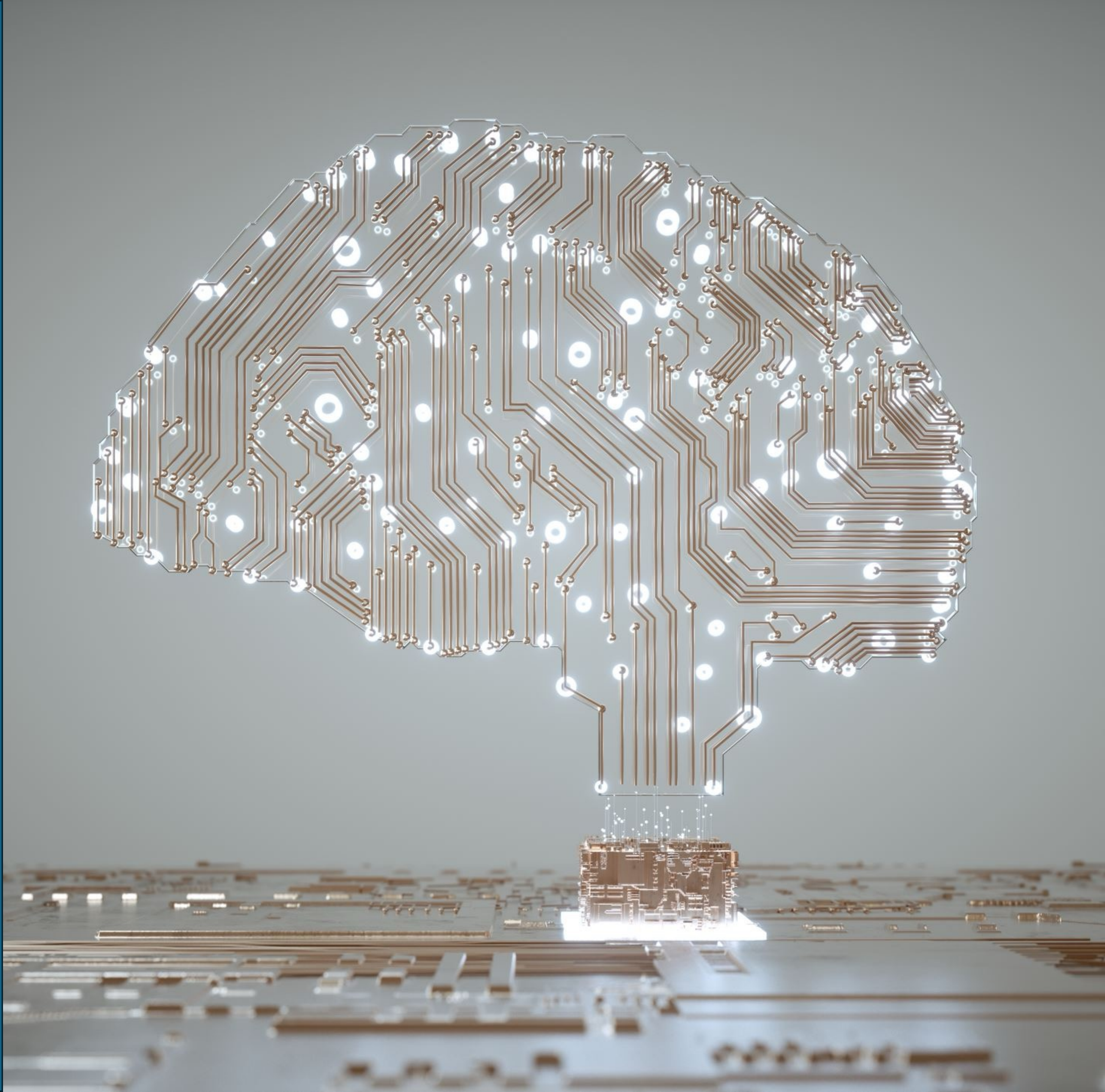


Overview: AI Challenges to IP Regimes

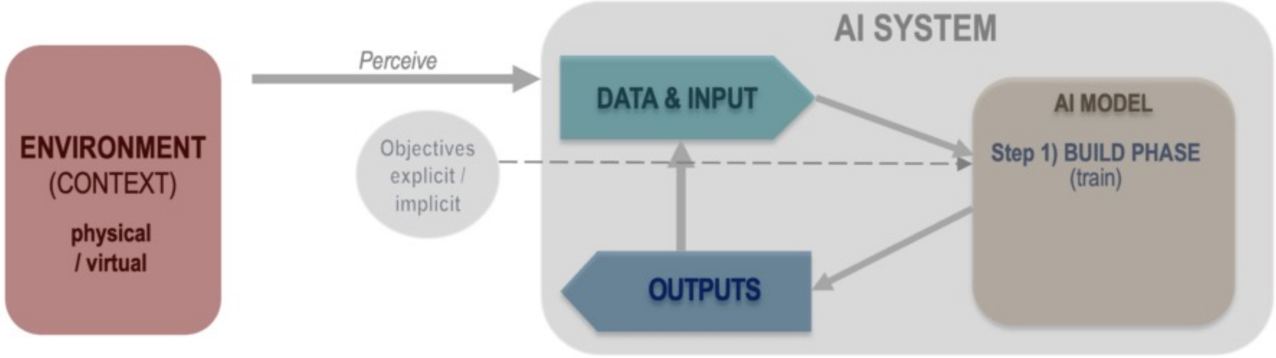
- How are AI systems protected by existing Intellectual Property (IP) regimes, i.e., patent and copyright?

- Current IP laws do not protect AI-generated works. Should that change? How should IP laws protect "AI-assisted creativity"?

- Training generative AI models requires large amounts of data, often sourced from the Internet. Does the training and usage of these models infringe on the rights of copyright holders?

Artificial Intelligence (AI) Systems

BUILD PHASE:
An AI system is a **machine-based** system, that



- for **explicit or implicit objectives**
- **infers**, from the **input** it receives
- How to **generate outputs** such as predictions, content, recommendations, or decisions

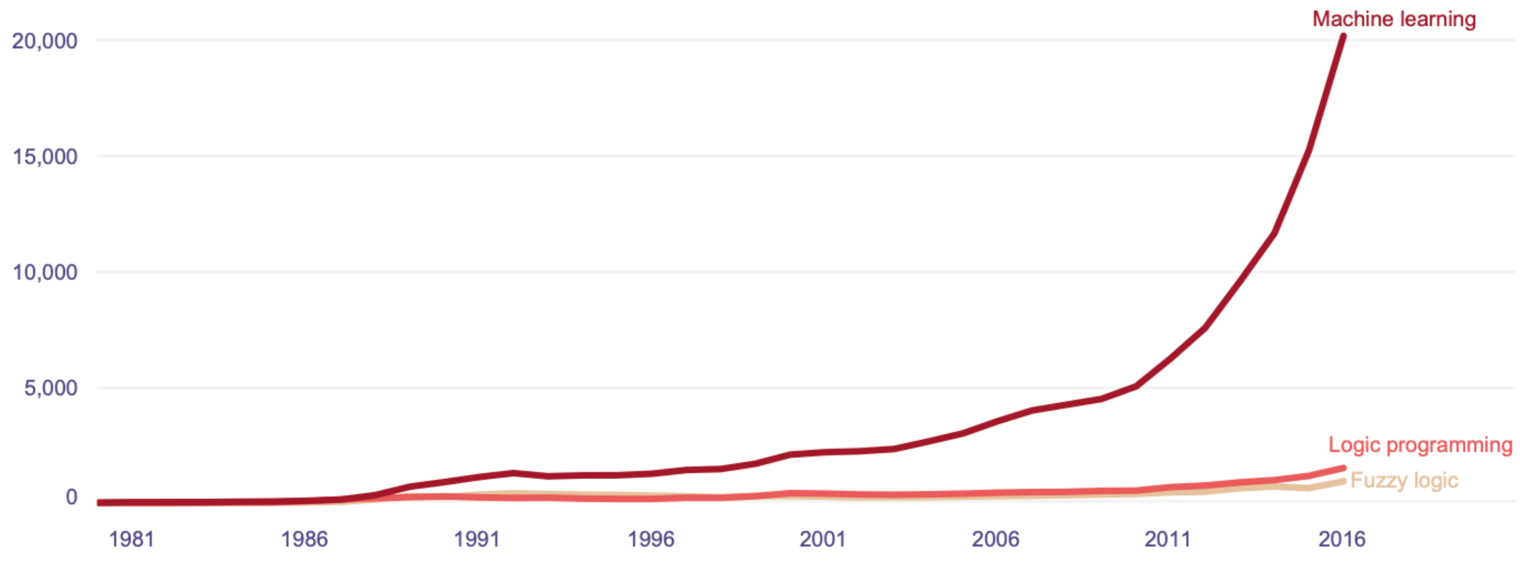
Source: OECD AI Policy Observatory

AI: "an evolving technology that simulates human intelligence processes using machines, especially computer systems"

→ Takes training data as input, learns algorithms to infer how to generate output

Rising Number of AI Patents

Figure 3.4. Patent families for top AI techniques by earliest priority year
Machine learning grew by an average of 26 percent annually between 2011 and 2016



Note: A patent may refer to more than one category

Source: WIPO Technology Trends 2019

Patent Protection for AI Systems

- AI-based systems e.g., deep learning, natural language processing, computer vision, speech recognition are AI/ML algorithms
- Algorithms: a set of steps or rules carried out by a software that takes data as input and produces an output to accomplish a specific task

→ abstract mathematical methods generally not eligible for patenting

But they can qualify for patenting if *applied in a practical way*, e.g., solve a technical problem, or improve an existing process & if they are *novel and involve an inventive step*

Copyright Protection for AI Systems

- Software copyright: legal protection for code meant to be read by a machine
- Protects "software source code": the way *source code* is written, but not the functionality of the software
- Software developers and companies use software copyright to prevent unauthorized copying, abuse, or exploitation of their software

Review: IP protection important to...

- **Inventors & Creators**

→Encourage them to innovate, by rewarding them with a fair return on their investments through rights to their own intellectual property

- **Businesses**

→IP assets are important strategic assets that preserve a company's competitive advantage: e.g., patents, industrial designs, software, etc.

Rethinking IP in the Age of AI

Key IP challenges in AI development and use:

- Current IP laws do not protect AI-generated works. Should that change? **Should IP laws protect "AI-assisted creativity"?**
- Training frontier AI models requires large amounts of data, often sourced from the Internet. **Does the training and usage of these models infringe copyright?**

Should IP Protect "AI creativity"?

Created by human?

Non-human entities:

- Monkey
- Machine

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- Monkey
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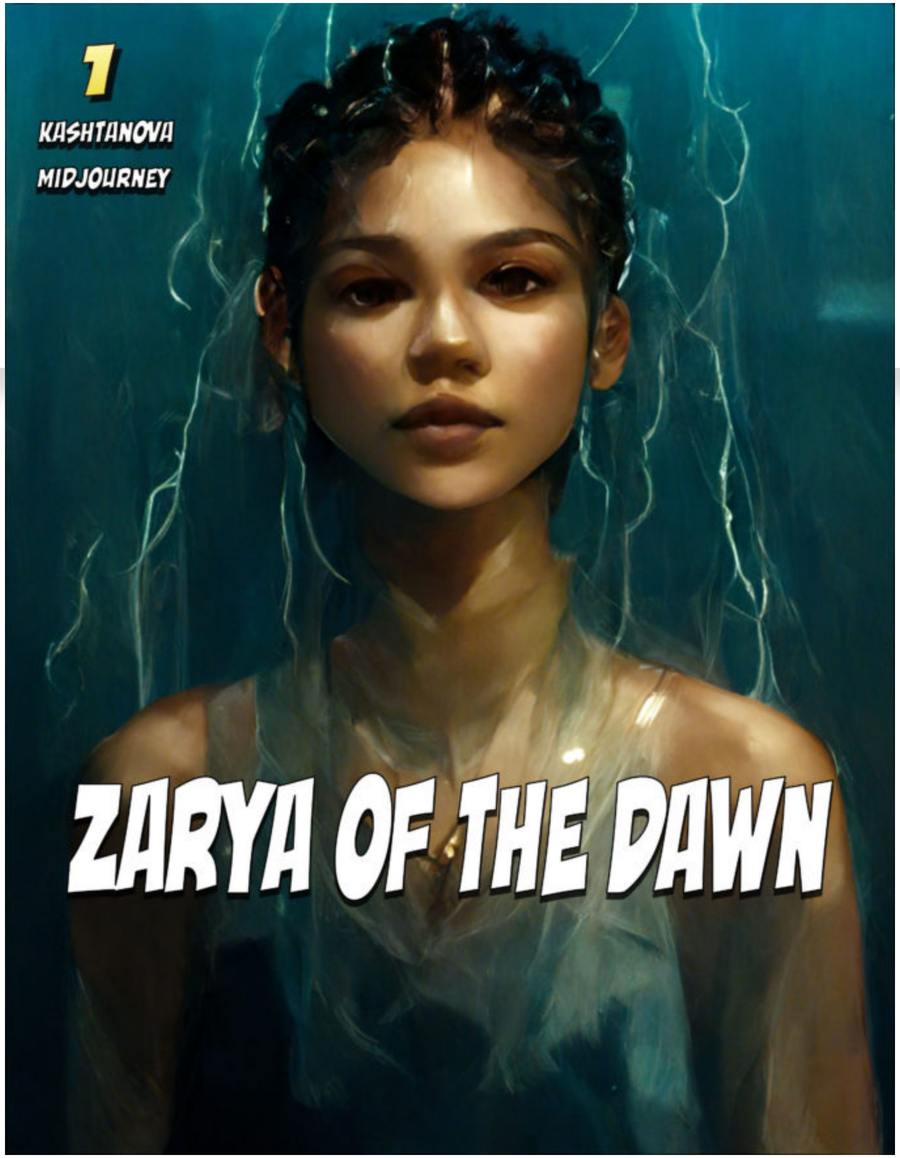
What if a human provided the prompts to a GenAI system to create texts, images, audios, videos?



United States Copyright Office

Library of Congress · 101 Independence Avenue SE · Washington DC 20559-6000 · www.copyright.gov

February 21, 2023

Van Lindberg
Taylor English Duma LLP
21750 Hardy Oak Boulevard #102
San Antonio, TX 78258

Previous Correspondence ID: 1-5GB561K

Re: Zarya of the Dawn (Registration # VAu001480196)

Dear Mr. Lindberg:

The United States Copyright Office has reviewed your letter dated November 21, 2022, responding to our letter to your client, Kristina Kashtanova, seeking additional information concerning the authorship of her work titled *Zarya of the Dawn* (the "Work"). Ms. Kashtanova had previously applied for and obtained a copyright registration for the Work, Registration # VAu001480196. We appreciate the information provided in your letter, including your description of the operation of the Midjourney's artificial intelligence ("AI") technology and how it was used by your client to create the Work.

The Office has completed its review of the Work's original registration application and deposit copy, as well as the relevant correspondence in the administrative record.[1] We conclude that Ms. Kashtanova is the author of the Work's text as well as the selection, coordination, and arrangement of the Work's written and visual elements. That authorship is protected by copyright. However, as discussed below, the images in the Work that were generated by the Midjourney technology are not the product of human authorship. Because the current registration for the Work does not disclaim its Midjourney-generated content, we intend to cancel the original certificate issued to Ms. Kashtanova and issue a new one covering only the expressive material that she created.

GenAI Can Mimic Human Creativity

- Existing AI systems are already powerful enough to mimic human creativity (e.g., passing the "Turing Test")

→ **GenAI tools** can create artworks of impressive quality that are hard to distinguish from works generated by human intelligence

Artistic works (images)	DALL-E 3, Midjourney, Stable Diffusion
Literary work (texts)	ChatGPT, Claude, Gemini
Music	Suno AI, AVIA, Udio
Code	GitHub Copilot, Replit Agent

Machine Creations Cannot Be IPs

- **Can an AI be considered the author of an intellectual property (IP)**, e.g., an artwork or an invention, under the current IP regime?

Machine Creations Cannot Be IPs

- **Can an AI be considered the author of an intellectual property (IP)**, e.g., an artwork or an invention, under the current IP regime?

No. Only human persons and organizations can.

→ Hence, AI cannot have ownership of an intellectual property (IP)

Machine Creations Cannot Be IPs

- **Can something created by an AI or with the aid of an AI be an intellectual property (IP)**, and hence protected by IP laws?

Under current IP regimes, introduced before the rise of AI:

- Economic and moral rights to the intellectual property can only be assigned to *legal personalities (e.g., humans and organizations)*
- *Not applicable to AI entities* as they do not have legal personality

Machine Creations Cannot Be IPs

- **Can something created by an AI or with the aid of an AI be an intellectual property (IP)**, and hence protected by IP laws?

Currently also *no*: current IP regimes protect ***human*** author(s) of an IP by assigning the IP's ownership to them, but AI is not human

→ *Other creative entities* such as an AI have not been considered

But IP Laws Should Address AI Creativity

- **AI creative capabilities** are developing fast, and **can potentially be highly valuable** → calls for rethinking the nature of creative work

- If AI is the creator of a valuable invention or creative work, <u>why can it not be credited as such</u> (as author of an otherwise legitimate IP)?

→ Existing legal framework is incomplete, which only protects human creativity, but does not consider creativity of non-human entities

What Likely Won't Work in the Long-Run

- **Deny IP protection to any work generated by AI systems** (i.e., maintain current status quo): Suboptimal because it discourages co-creative experimentation involving the use of an AI system

- **Making AI legal persons and assigning IP rights to the AI itself**: Not a very realistic approach, since AI cannot act on its own will (for now); and if it ever does, we'll face whole new types of risks of a sentient AI, which are far beyond addressing IP concerns

Update IP Laws to Incorporate AI: How?

- IP laws should be changed to account for **AI-generated or AI-assisted** creative works
- AI-assisted creations have both human and AI inputs, and both human and AI are involved in the creative process
- Results of the activity of AI system can depend on human input or choices made by human operators, to varying degrees

Who Should Be Author(s) / Owner(s)?

- Person who *made the necessary arrangement for the AI to create the work* (by interacting with the AI system, e.g., prompting a GenAI app)?

→ This is the case e.g., in the UK, for determining IP ownership

- Other alternatives: What about the ***person(s) who invented or set up the AI system***? What about the ***owner(s) of the AI system***?

Decisions to designate IP rights to any author(s) should also depend on:

- Degree of AI involvement and perceived worth of the final output

Rethinking IP in the Age of AI

Key IP challenges in AI development and use:

- Current IP laws do not protect AI-generated works. Should that change? **Should IP laws protect "AI-assisted creativity"?**

- Training frontier AI models requires large amounts of data, often sourced from the Internet. **Does the training and usage of these models infringe copyright?**

Can Copyright Law Stop Generative AI?

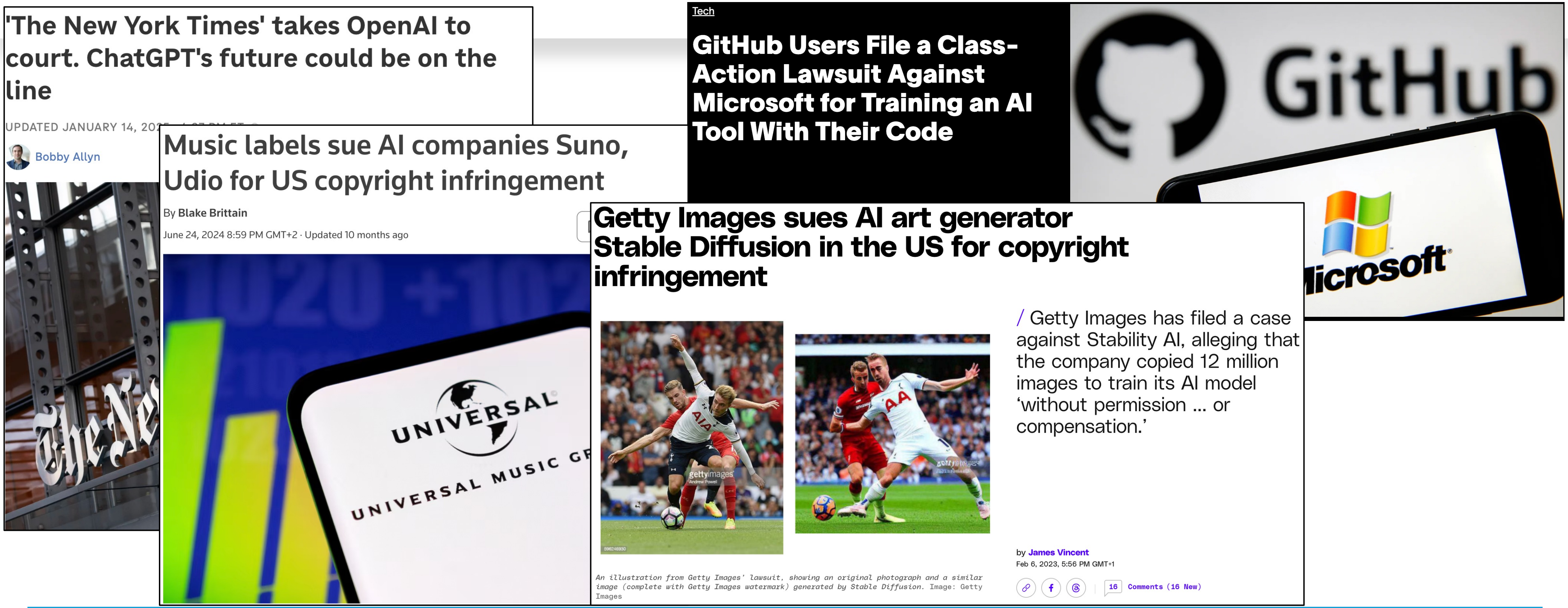
Source: https://www.youtube.com/watch?v=H3fqEW1nwpQ

Can Copyright Law Stop Generative AI?

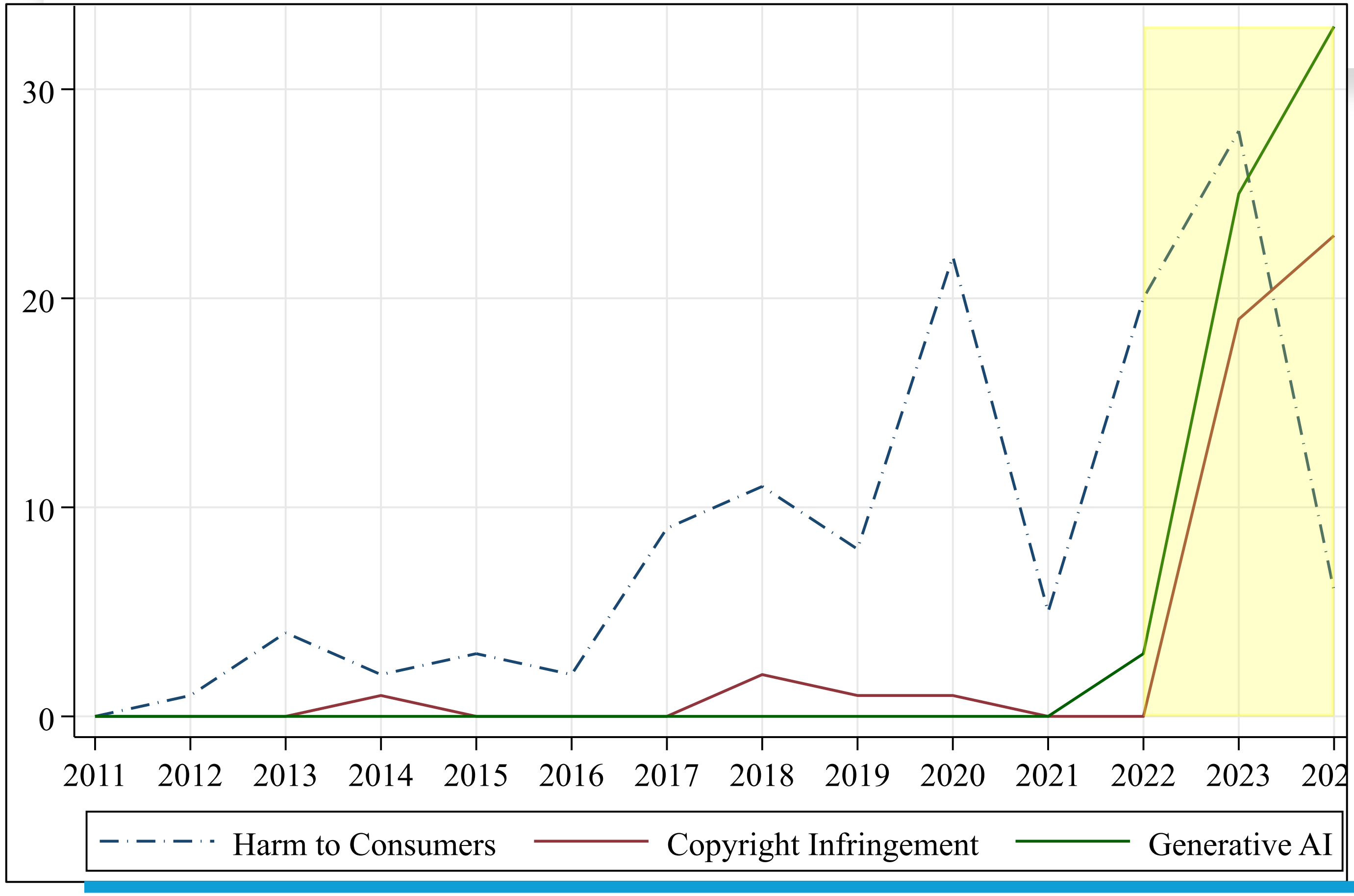
- Creators and artists are worried about their works being used to train AI without their permission, and that AI will displace their jobs
- Copyright law can potentially stop or slow down GenAI development, but there are pros and cons
- Resolution of recent GenAI lawsuits will have a profound influence on the future landscape of AI development

Source: https://www.youtube.com/watch?v=H3fqEW1nwpQ

Generative AI Copyright Lawsuits



Sharp Rise in AI Lawsuits Since 2022



- Prior to 2022, most lawsuits are around consumer harms due to AI, e.g., algorithmic bias and discrimination, privacy violations, inciting violence
- Since 2022, rapid rise of lawsuits related to copyright infringement and generative AI

Source: https://blogs.gwu.edu/law-eti/ai-litigation-database-search/

Record Labels vs. AI Music Generators

Reflection questions:

- Who are the defendants and plaintiffs involved in the case? What did they do?
- What is the plaintiffs' main claim?
- What is the defendants' main defense?
- What do you think is the presenter's view, and what is your view?

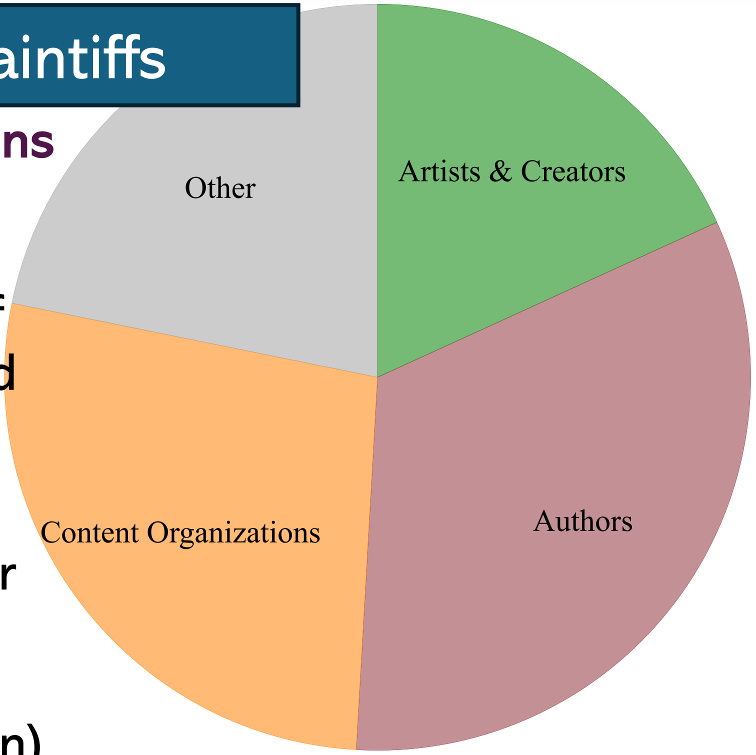
Note terminology: "transformative", "fair use", "consent, compensation, and credit"

Source: https://www.youtube.com/watch?v=Hy8Or6af98c

GenAI Copyright: Who is Suing Whom?

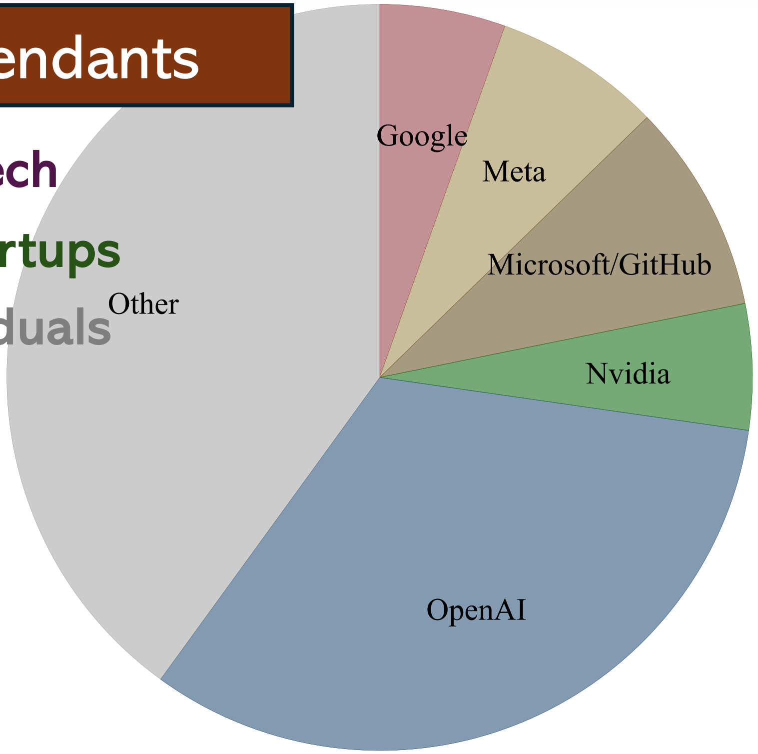
Plaintiffs

- **Corporations** that own large amounts of copyrighted content
- **Individual creators**, or groups of creators (class-action)



Defendants

- **Big Tech**
- **AI Startups**
- **Individuals**



Source: https://blogs.gwu.edu/law-eti/ai-litigation-database-search/

Another High-Profile Lawsuit...

The New York Times is suing OpenAI and Microsoft for copyright infringement

Photo by Kena Betancur/VIEWpress

/ A lawsuit claims OpenAI copied millions of Times' articles to train the language models that power ChatGPT and Microsoft Copilot.

by **Emma Roth**
Dec 27, 2023, 2:49 PM GMT+1

126 Comments (126 New)

- Latest: Federal judge ruled that the copyright lawsuit can proceed, keeping most but not all of the plaintiffs' claims (in March 2025)
- The outcome will shape the trajectory of GenAI development, setting a precedent for similar cases

GenAI Copyright Lawsuits in a Nutshell

Two lines of claims:

- **Input (training data)**

→ Defendant ingest large amounts of plaintiffs' works as input data for training its generative AI models

- **Output (AI-created works)**

→ Defendant's generative AI system produced (in response to user prompts) derivative outputs that infringe on plaintiffs' original works

→ Other harms e.g., hallucination and misinformation attributed to plaintiffs, causing brand reputation damage and trademark dilution

GenAI Copyright Lawsuits: Training Data

Key ingredient to powerful GenAI is **high-quality training data**

- Ingests tremendous amounts of data, often containing copyrighted materials, to train state-of-the-art algorithms

- Input data quality especially important for model performance
  - NYTimes is a major data source for training OpenAI's models, and given larger weights than many other data sources due to its high quality
  - Other high-quality sources like Common Crawl (4th biggest content corpus), WebText2 (containing NYTimes) are given very heavy weight
  - Image generation models primarily trained on LAION-5B

GenAI Copyright Lawsuits: AI Outputs

AI-generated outputs can [mimic and compete with human works]

- Some prompted AI outputs are sufficiently similar to and directly infringe on copyrighted content
- AI-generated works can compete with the original works on which the generative models were trained (lowering demand and prices)
- Many creative professionals (e.g., artists, musicians, authors) are concerned that generative AI may displace their jobs and driven down wages, and eventually transform the entire profession

Defendant's Main Lines of Defense

Under US law, fair use of copyrighted materials do not infringe copyright

- Defendants can claim their use of copyrighted materials to train AI models constitute "fair use"
- Key factors for establishing fair use: *purpose of the use*; its *effect on the "market for or value of the copyrighted work"* (Samuelson, 2023)
- **"Transformative use"** of copyrighted work → The purpose of using these materials to develop the GenAI system is to ***generate completely new outputs*** instead of memorizing and regurgitating existing content

History of Innovation vs. IP Protection

- Innovations in digital technologies are historically associated with copyright controversies
  - For example, *Authors Guild v. Google* ruled in favor of Google in 2015: Google Books raised the demand for physical books by increasing the chance of discovery by consumers (Nagaraj & Reimers, 2023)

- ***Is it different this time?***

History of Innovation vs. IP Protection

- The goal of copyright law has been to “foster the creation and dissemination of knowledge for the public good”
- Not based on fundamental rights, but based on societal and economic considerations, to strike a balance between interests of:
  - Copyright holders: To prevent misappropriations of their works that undermine the incentives to create
  - Follow-on innovators: To provide them with sufficient space to develop follow-on innovations or technologies

Challenges to Plaintiffs’ Claims

<u>Getting concrete evidence of infringement may prove difficult</u>

- Training data are not exactly “copies”, but they are tokenized mathematical representations of the original content
- Copyright law protects only the original expression (e.g., “sequences of words” or “melody of music”), not ideas, facts, methods, elements, underlying subjects, or patterns of connections among concepts (Samuelson, 2023)
- AI companies claim “transformative” use with the purpose of generating entirely new works instead of replicating the original

Challenges to Plaintiffs' Claims

Training data may come from non-infringing sources

- Training data may not be directly collected by the plaintiffs
- Most image GenAI systems used LAION-5B for training – a free public dataset with 6 billion images linked to text collected from the Internet by the German research nonprofit LAION
- LAION-5B is created lawfully due to EU's exception for nonprofits to use copyrighted works for TDM (text and data-mining), and commercial actors to engage in TDM (copyright owners can opt out) because the data can be used to create new knowledge

Challenges to Plaintiffs' Claims

AI developers can claim "transformative use" of copyrighted works

- This is important for establishing ***fair use***, to counter the plaintiffs' claim that the defendant's use of copyrighted materials is "non-transformative and commercial"
- Transformative: "To add something new, with a further purpose or different character, & do not substitute for the original use of the work"
- ***Transformative use*** purpose is favored in establishing ***fair use***

Challenges to Plaintiffs' Claims

Courts may consider public benefits and creative impacts to society

- In weighing ***market effects***, courts may consider wider market-level impact, since copyright law aims to promote creative progress (regardless of the innovator's identity), not only the plaintiffs' revenues
- Open-source AI model developers can claim societal benefits: enabling follow-on innovations & lowering costs to productive use of AI outputs
- Revenue loss easier to justify for corporations (e.g., clearly defined licensing market where GenAI systems reduces licensing revenues of copyright owners) than for an individual whose work is marginal

What Happens If AI Companies Lose?

How will AI companies respond if the plaintiffs' claims prevail?

Immediate consequences:

- Remove copyrighted works from training data, and only train on public domain and open license content → Result in less capable AI systems
- GenAI companies will seek direct licensing from copyright holders and marketplaces to acquire training data → Larger and more resourceful firms are more likely to afford them and maybe get better deals
- Not only the defendant in the case but other companies developing GenAI systems will also be affected, since it will be used as a precedent

What Happens If AI Companies Lose?

How will AI companies respond if the plaintiffs' claims prevail?

Long-run consequences (downstream or unintended consequences):

- Generative AI developers may move their operations to countries with slacker copyright law enforcement
  - Countries farther from the technology frontier have particularly large incentives to encourage "pirating", to spur follow-on innovation at the expense of the original inventor (e.g., see Moser's book)
- Large technology firms may dominate generative AI development
  - Licensing costs much more cumbersome for smaller firms and startups
- Non-commercial entities (e.g., academic researchers) may become more dependent on large tech firms' open-source projects (if they continue)

Limitations of Current Copyright Law…

- Imagine if the AI models get so good that the quality of their outputs are judged to be on par with or better than the works produced by human creators, and
- "Transformative use" holds – outputs never infringe directly on copyrighted works (e.g., output filters to prevent duplicates)

→ May still harm original content creators – who were not credited or compensated, and did not have control over how their works were used to train AI

Is "Fair Use" Sufficient?

- "Fair use" allows *use of copyrighted materials without requiring permission from the rights holder*, and can be granted if the use is deemed transformative, and creates value for the market

- But **this does not seem fair to copyright owners** whose works were used to train the AI for free

Is “Fair Use” Sufficient?

- Should explicitly consider the original content providers

- Proposal (Gans, 2024): should incorporate the tradeoff between

1) The level of potential harm to original content providers, and
2) The importance of content for AI training quality

Is "Fair Use" Sufficient?

Determining potential harm to original content providers

- If the AI simply does what humans do: e.g., summarizing various news articles from different sources to produce a news report, it should be covered by fair use provisions (just like news reporters doing their job)
- But the AI system may do other things that cause commercial damages to the copyright holder: job displacement, brand dilution, reputation damage, etc.

Is “Fair Use” Sufficient?

Determining the importance of the content for AI training quality

- Scale matters (Gans, 2024): cannot trace the “provenance” (origin and history of a piece of content) in large-scale GenAI systems, thus it is difficult to know the source of the harm ahead of time

- Proposed solution: “ex-post fair use assessment”

- Licensing of training data: require large AI developers to pay for copyright owners’ permission for using their content as training data

Labor Market Impact of GenAI?

Not only content owners, but many creative professionals (e.g., artists, writers, voice actors, musicians, online creators) are strongly opposed to their works being used to train AI

→ This is rooted in the fear that they will be replaced by machines (these concerns are based on economics rather than fundamental rights)

Empirical evidence seems important:

1) How does the introduction of GenAI systems affect labor markets?
2) What do people use GenAI systems for?

Labor Market Impact of GenAI?

- Classical framework is “substitution vs augmentation”: does GenAI **substitute** or **augment** humans in conducting a specific job task?

→ Not always clear, and the nature of a job task can change over time

- Literature measures each occupation’s “exposure” to AI (Felten et al., 2021), which can be substitution, augmentation, or both
- Some initial evidence that the introduction of a GenAI (e.g., ChatGPT) reduced employment and earnings of online freelancers (Hui et al., 2024), and reduce contribution to online knowledge community (Burtch et al., 2024)

→ But we don’t know much about long-run impact and more profound changes over time

Labor Market Impact of GenAI?

Not really much evidence on creative professionals specifically yet…

The consequences of generative AI for online knowledge communities

Gordon Burtch, Dokyun Lee & Zhichen Chen

Scientific Reports **14**, Article number: 10413 (2024) | Cite this article

Abstract

Generative artificial intelligence technologies, especially large lang
ChatGPT, are revolutionizing information acquisition and content p
of domains. These technologies have a significant potential to imp
content production in online knowledge communities. We provide
analyzing data from Stack Overflow and Reddit developer commu
2021 and March 2023, documenting ChatGPT's influence on user
observe significant declines in both website visits and question vo
particularly around topics where ChatGPT excels. By contrast, acti
shows no evidence of decline, suggesting the importance of socia
the community-degrading effects of LLMs. Finally, the decline in participation on Stack Overflow is found to be concentrated among newer users, indicating that more junior, less socially embedded users are particularly likely to exit.

The Short-Term Effects of Generative Artificial Intelligence on Employment: Evidence from an Online Labor Market

Xiang Hui, Oren Reshef, Luofeng Zhou

Published Online: 24 Sep 2024 | https://doi.org/10.1287/orsc.2023.18441

Abstract

Generative artificial intelligence (AI) holds the potential to either compleme
their productivity or substitute them. We examine the short-term effects of
generative AI models (ChatGPT, DALL-E 2, and Midjourney) on the employm
freelancers on a large online platform. We find that freelancers in highly af
from the introduction of generative AI, experiencing reductions in both em
find similar effects studying the release of other image-based generative A
heterogeneity by freelancers' employment history, we do not find evidence
measured by their past performance and employment, moderates the adve
employment. In fact, we find suggestive evidence that top freelancers are d
by AI. These results suggest that generative AI may transform the role of h
organization and reduce overall demand for workers.

Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses

Edward Felten, Manav Raj, Robert Seamans

First published: 28 April 2021 | https://doi.org/10.1002/smj.3286 | Citations: 32

SECTIONS PDF TOOLS SHARE

Abstract

Research Summary

We create and validate a new measure of an occupation's exposure to AI that we call the AI Occupational Exposure (AIOE). We use the AIOE to construct a measure of AI exposure at the industry level, which we call the AI Industry Exposure (AIIE) and a measure of AI exposure at the county level, which we call the AI Geographic Exposure (AIGE). We also describe several ways in which the AIOE can be used to create firm level measures of AI exposure. We validate the measures and describe how they can be used in different applications by management, organization and strategy scholars.

Sticky Laws and Changing Technologies

Laws need to keep up with the rapid progress of AI technologies

- To **incentivize the development and use** of AI and **complementary value creation** activities, but also to ensure sufficient **protection against potential side-effects** of AI systems
- Creators can obtain fair returns to their creative work, get credit for their work, share the results with society, and control the transfer of their IP rights

Homework Assignment

Choose one of the following two options

- Present an in-depth analysis of an open-source LLM provider of your choice:
  - What are its best models, and what are the licensing types of these models? How can one use these models?
  - What are the advantages (and disadvantages) of this provider relative to its key competitors? (hint: many angles e.g., country regulation, GPU resources, financial resources, existing users)
- Analyse a copyright infringement lawsuit around Generative AI:
  - Choose a case from DAIL: https://blogs.gwu.edu/law-eti/ai-litigation-database-search/
  - What is the current status of the lawsuit (list ALL recent updates)?
  - If the suit is not yet settled, do you think the plaintiff/defendant is going to win? Why?
  - If the suit is not yet settled, how will you rule as a judge? Why?

Homework Assignment

More suggestions on the second option:

- Analyse a copyright infringement lawsuit around Generative AI:
  - Choose a case from DAIL: https://blogs.gwu.edu/law-eti/ai-litigation-database-search/
  - Or you can analyse a case of your own choice (preferably one where the plaintiffs are based in Europe)
  - Explore different angles, not in abstract but with concrete connections to the case:
    - Legal: “fair use” key criteria (1) purpose of the use and (2) impact on the market (Samuelson, 2023)
    - Economic: “public good” market-level consequences for innovation, firm revenue, price, competition, etc.
    - Ethical: how can we ensure fairness to original creators, without slowing technical progress or increasing large firms’ entrenchment?
  - [Optional] You can borrow ideas from the papers in the required & optional readings on Canvas, or assemble your own literature to guide your analyses

References

- Atik, J., Wahlberg, L., Jeutner, V., Selberg, N., Andersson, U., Mattsson, T., Nordberg, A., Nowag, J., & Persson, V. (n.d.). *AI & Law* [MOOC]. https://www.coursera.org/learn/ai-law
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